

JEREMY ACHONG

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EDUCATION

University of Central Florida Orlando, FL

December 2026

Bachelor of Science in Computer Engineering

- Relevant course work: Data Structures & Algorithms, Discrete Mathematics, Operating Systems, Computer Architecture, Object-Oriented Programming, Digital Logic Design, Computer Vision, Real-Time Systems

TECHNICAL SKILLS

- Python, Java, C/C++, TypeScript, SQL, REST APIs, Git, Linux, Agile Workflows

WORK EXPERIENCE

Software Developer

Siemens Energy- Orlando, FL

May 2025 - Present

- Developed multi-language software modules (**Java, Python, C, Fortran**) to parse and process FEA results, accelerating complex lifecycle calculations for turbine material fatigue testing and life-prediction
- Enhanced internal tool by migrating file formats to HDF5, reducing storage footprint by 90% and achieving a 2-millisecond read latency to accelerate data pipelines for global engineering groups
- Partnered directly with component design groups to improve the **PyView** based 3D viewer with new 10+ user centered features, including automatic report generation
- Tracked user permissions and program usage with **MySQL** database to gain insight into 100+ person expert group
- Automated legacy deployments by optimizing **Bash** scripts into modern, **Git**-triggered **CI/CD** pipelines for 4 applications for Linux environment

Independent Contractor

Freelance Website Development- Orlando, FL

September 2024 – Present

- Partnered with business owners on curating brand identities and web storefronts, boosting online presence to generate an increase of customer reach
- Utilized **Figma** to prototype website sitemap and architected custom **Next.js** applications tailored for complex, high-performance web requirements
- Engineered custom **JavaScript, HTML, and CSS** to bypass Webflow limitations, revamping the user interface to seamlessly integrate location-based features across three venues for client

Library Technology Student Assistant

University of Central Florida Library- Orlando, FL

August 2022 - September 2024

- Administered the daily checkout and intake of diverse technical hardware, meticulously inspecting laptops, A/V gear, and peripherals to ensure optimal functionality for student patron use
- Monitored and resolved 20+ weekly hardware/software issues across Windows, Mac, and Linux, escalating critical system failures and access control problems to senior staff via Trello's ticketing system

Lighting Technician

Northland Church- Longwood, FL

January 2017 - September 2022

- Programmed live lighting sequences using grandMA2 software to support immersive show experiences
- Collaborated with multi-functional Audio and Video teams to craft cohesive live experiences and resolve technical issues across software and hardware interfaces

- Maintained 200+ light fixtures and addressed technical issues such as **DMX** network patching, broken shutters, and failed lamps to ensure efficient and reliable system operations

PROJECT LIST

Pillba - Automated Medication Dispenser | IEEE Elevate Grant Recipient June 2025 - Present

- Employed Behavior Driven Design (BDD) to define acceptance criteria for 20+ accessibility and hardware features, ensuring the system's workflow directly aligned with the specific needs of elderly and disabled users.
Designed an inclusive high accessibility UI, with **PyQt** adhering to WCAG guidelines
- Leveraged **Tesseract OCR** to automate prescription bottle text extraction, eliminating manual data entry and simplifying user workflow for curved-text processing
- Engineered low-level device drivers for real-time motor control and data acquisition from IR, load cell, and Hall effect sensors for accurate medication dosage dispensing
- Constructed **SQLite** database to store captured drug adherence statistics and prescription information

Travel Genie - AI Powered Trip Planner January 2025 - April 2025

- Designed a full-stack application integrating Gemini models via REST APIs to generate personalized user travel plans.
- Deployed application to both web and mobile platforms, leveraging **React** and **Flutter** to ensure a seamless cross-platform experience
- Led 6 developers in architecting core application features from ideation to deployment, managing project schedules via Gantt charts

LEADERSHIP EXPERIENCE

Web Design Committee Member September 2025 - Present

IEEE at UCF

- Engineered an account dashboard feature for the IEEE UCF website, integrating QR code generation to streamline event check-in process using **Next.js**

Media Committee August 2022 - February 2025

Vietnamese American Student Association (VASA)

- Produced a multi-camera livestream setup using **OBS Studio**, **RTMP streaming**, and DSLR/webcam feeds, configured encoder settings (bitrate, resolution) to ensure low-latency, high-quality, 4k video broadcasts